Synchronous Communications

In synchronous communication, services interact in a *request–response* manner — the caller sends a request and waits for a reply before continuing. This model is simple and intuitive, resembling traditional client-server interactions. Typical implementations include **REST**, **GraphQL**, **gRPC**, **Avro**, each offering different trade-offs in flexibility, performance, and data format (best for edge or cloud).

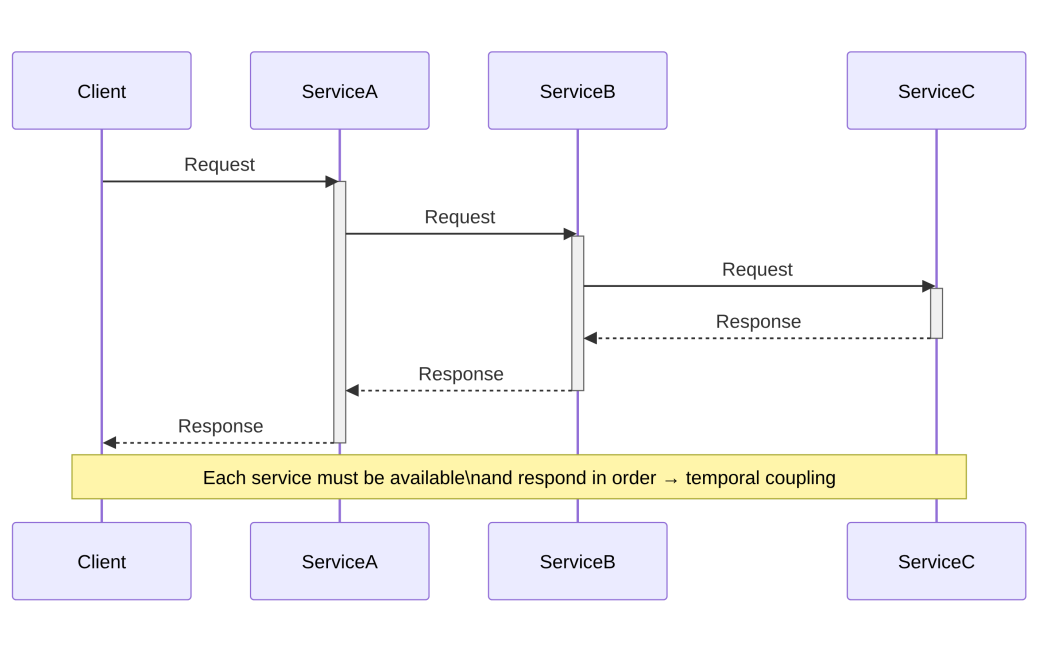
Definitions and Known Issues

Spatial coupling refers to the degree of dependency between different components or services in a system at a given point in time. A high degree of spatial coupling means that components are tightly connected, requiring direct knowledge of each other's existence, interfaces, or locations. This can lead to reduced flexibility and increased maintenance complexity.

```java
public class Payment {
    private double amount;
    private String paymentMethod;
    private boolean isApproved;
}
```

```java
@RestController
public class OrderController {

    public Order createOrder(Order order) {
        // Directly calling Payment Service location, port, endpoint (tightly coupled)
        String url = "http://payment-service:8080/api/payments";
        RestClient restClient = RestClient.builder().build();
        Payment payment = restClient.get()
                .uri(url)
                .retrieve()
                // Direct deserialization (tightly coupled)
                .body(new ParameterizedTypeReference<Payment>() {});
        // ...
    }
}
```

Temporal coupling occurs when components or services must be available and responsive at the same time to function correctly. This often happens in synchronous communication patterns, where one component must wait for another to process a request before proceeding. In cases where a chain of multiple services need to communicate, **the cumulated latency can significantly degrade performance**. **[unsolvable with synchronous approaches!]**


Client
ServiceA
ServiceB
ServiceC
Request
Request
Request
Response
Response
Response
Each service must be available\nand respond in order → temporal coupling
Client
ServiceA
ServiceB
ServiceC

API coupling refers to the degree of dependency between a client and an API. A highly coupled API means that changes in the API can easily break the client, while a loosely coupled API provides more flexibility and resilience to changes.

Example: Modifying the `Payment` class:

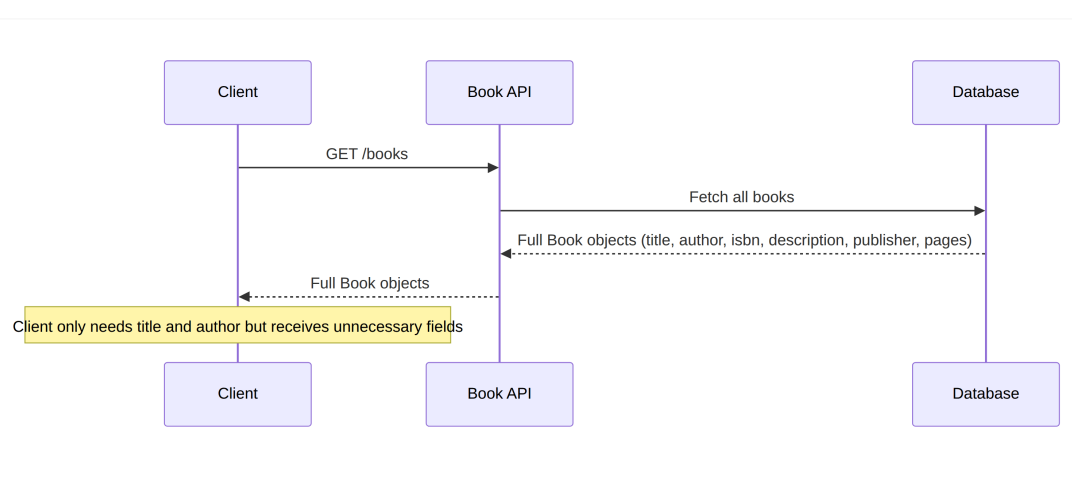
```java
public class Payment {
    private double amount;
    private String currency;
    private String paymentMethod;
    private boolean isApproved;
}
```

Over-fetching occurs when an API returns more data than the client actually needs, leading to wasted bandwidth and increased serialization time. This typically happens in REST APIs with fixed response structures, where a client cannot specify exactly which fields it requires.

Scenario: A client wants only the title and author of a book, but the API returns the entire book object, including unnecessary fields like ISBN, description, publisher, etc.

```java
public class Book {
    private String title;
    private String author;
    private String isbn;
    private String description;
    private String publisher;
    private int pages;

    // ...
}
```

```java
@RestController
@RequestMapping("/books")
public class BookController {

    @GetMapping
    public Iterable<Book> findAll() {
        return List.of(
            new Book("Spring Boot", "John Doe", "123456789", "Comprehensive guide", "TechPub", 500),
            new Book("Microservices", "Jane Smith", "987654321", "Detailed explanation", "CloudPub", 300)
        );
    }
}
```

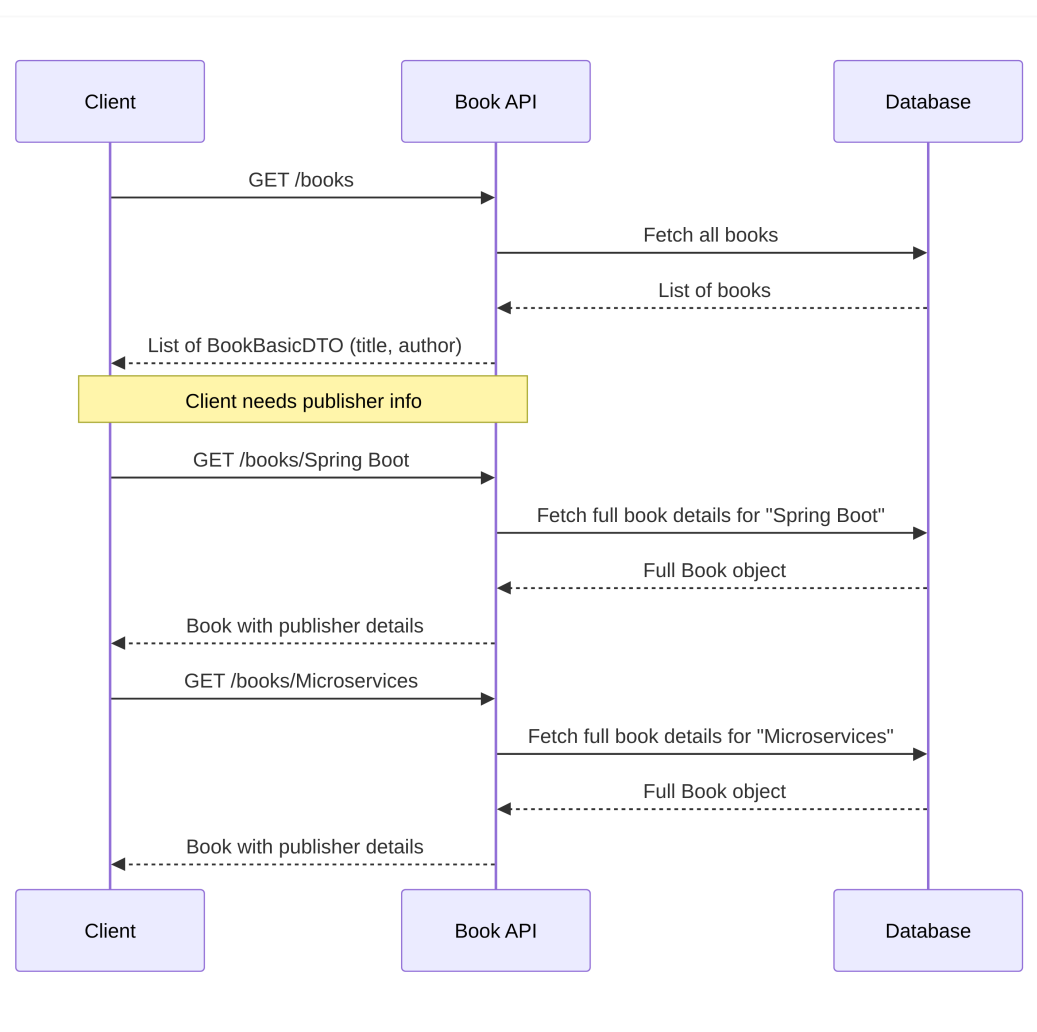

Client
Book API
Database
GET /books
Fetch all books
Full Book objects (title, author, isbn, description, publisher, pages)
Full Book objects
Client only needs title and author but receives unnecessary fields
Client
Book API
Database

Under-fetching (aka chattiness) occurs when a client requests data from an API but does not receive all the necessary information in a single response. As a result, the client must make additional requests to retrieve the missing data, leading to inefficiencies, increased latency and costs.

Scenario: The endpoint `/books` returns a simplified model for books. If the client also needs the publisher, it must make additional requests to fetch publisher details such as: `GET /books/{title}`

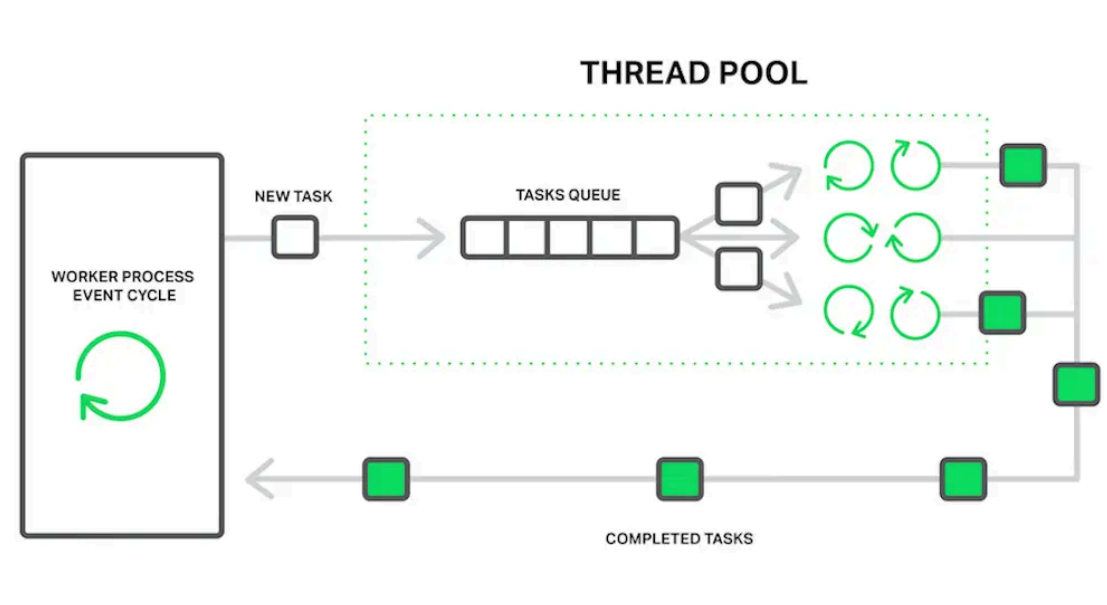
```java
public class BookBasicDTO {
    private String title;
    private String author;
}
```

```java
@RestController
@RequestMapping("/books")
public class BookController {

    @GetMapping
    public Iterable<BookBasicDTO> findAll() {
        List<Book> books = List.of(
                new Book("Spring Boot", "John Doe", "123456789", "Comprehensive guide", "TechPub", 500),
                new Book("Microservices", "Jane Smith", "987654321", "Detailed explanation", "CloudPub", 
300)
        );

        // Returns only title and author
        return books.stream()
                .map(book -> new BookBasicDTO(book.getTitle(), book.getAuthor()))
                .toList();
    }
}
```


Client
Book API
Database
GET /books
Fetch all books
List of books
List of BookBasicDTO (title, author)
Client needs publisher info
GET /books/Spring Boot
Fetch full book details for "Spring Boot"
Full Book object
Book with publisher details
GET /books/Microservices
Fetch full book details for "Microservices"
Full Book object
Book with publisher details
Client
Book API
Database

Thread pool exhaustion (on client side): Clients waiting for a response from the server consume system resources (threads, memory), which can be problematic in high-throughput environments. **[unsolvable with synchronous approaches!]**

Scenario: A client queries the endpoint `/books` 100 times every second. Each request takes on average 1 second to complete. At any moment, the **client** has approximately 100 threads in a waiting state. Given that each thread requires 2MB of RAM, what happens if the books service stops responding for 10 seconds? The client will need ~2GB of RAM just to manage waiting threads.


THREAD POOL
WORKER PROCESS EVENT CYCLE
NEW TASK
TASKS QUEUE
COMPLETED TASKS

REST (JSON over HTTP) Alternatives

Limitations of REST:

Characteristic	REST
Serialization	Y
Temporal Coupling	Y
API Coupling	Y
Over-fetching	Y
Under-fetching (chattiness)	Y
Thread Pool Exhaustion	Y

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gRPC (Protobuf over gRPC) / Avro

- **Serialization**: Compact binary serialization
- **API Coupling**: Support backward/forward compatibility.
- **Over-fetching**: Efficient binary serialization reduces unnecessary data.
- **Under-fetching (chattiness)**: HTTP/2 multiplexing minimizes multiple round trips.

Characteristic	REST	gRPC
Serialization	Y	Reduced
Temporal Coupling	Y	Y
API Coupling	Y	Reduced
Over-fetching	Y	Reduced
Under-fetching (chattiness)	Y	Reduced
Thread Pool Exhaustion	Y	Y

GraphQL

- **Serialization**: JSON responses include only the fields requested by the client.
- **API Coupling**: Schema evolution supports backward and forward compatibility.
- **Over-fetching**: Clients fetch exactly the data they need, avoiding unnecessary fields.
- **Under-fetching (chattiness)**: Nested queries allow fetching multiple related resources in a single request.

Characteristic	REST	GraphQL
Serialization	Y	Reduced
Temporal Coupling	Y	Y
API Coupling	Y	Reduced
Over-fetching	Y	Reduced
Under-fetching (chattiness)	Y	Reduced
Thread Pool Exhaustion	Y	Y

Protocols in the Edge-Cloud Continuum

- **REST (JSON/HTTP)**: simple, widely supported, cloud-friendly; higher latency limits edge suitability
- **GraphQL**: flexible queries, reduces over/under-fetching; JSON parsing may add latency at edge
- **gRPC (Protobuf)**: low-latency, binary, streaming; great for IoT, edge, real-time pipelines
- **Avro**: schema-based, compact binary; excellent for streaming, edge-cloud messaging, schema evolution

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Resources